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# MAE Preliminary Examination

Mathematics Section

Monday, Monday April 15, 2024, 9:00am-11:30noon

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|-----------|
| Your Name |
|-----------|

| THREE PROBLEMS WILL BE GRADED                      |        |
|----------------------------------------------------|--------|
| Select the 3 problems you've worked, to be graded: | points |
| Problem:                                           | /10    |
| Problem:                                           | /10    |
| Problem:                                           | /10    |
| Total                                              | /30    |

Please give your answers/work in the space provided

Explain your work/steps clearly

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**Linear Algebra: Problem 1 [10 points]:**

Consider the function of an angle  $\theta \in [-\pi, \pi]$  given by

$$f(\theta) = \left\| \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\|.$$

The norm is the Euclidean norm. Determine the minimal value of  $f(\theta)$ ,

$$f_{\min} = \min f(\theta),$$

as well as a value for the angle  $\theta_{\min}$  that achieves the minimal value for  $f(\theta)$ .

You can give the answer in a form that can be readily evaluated with a calculator, if you don't have one. I.e., an answer in the form that involves roots, and inverse trigonometric functions is acceptable.

Workspace for Problem 1: Explain your reasoning/work here.

**Solution:**

The matrix  $\begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix}$  is an orthogonal matrix, effecting a clockwise rotation on vectors  $\begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2$ .

**Solution approach A:** Consider  $f(\theta)^2$  since then, we need to consider and minimize

$$\begin{aligned} \left\| \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\|^2 &= 5 + 2 + 2 \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ &= 7 + 2(3\cos(\theta) - \sin(\theta)). \end{aligned}$$

To see the above note that  $\|x + y\|^2 = \|x\|^2 + \|y\|^2 + 2x^T y$ . It is simple to see that the minimizing angle must satisfy

$$-3\sin(\theta_{\min}) - \cos(\theta_{\min}) = 0,$$

The solution is  $\theta_{\min} = -0.3218 + k\pi$ . A simple drawing shows that a small negative angle lines up the two vectors,  $[2, 1]$  and  $[1, 1]$ , but this is not what we want. We want them to be co-linear and of opposite direction. Hence, the correct solution is

$$\theta_{\min} = -0.3218 + \pi \simeq 2.8[\text{rad}]$$

Then the minimal value for  $f$  will be when the two vectors are colinear and of opposite directions, and it will be the difference of their two lengths,  $f_{\min} = \sqrt{5} - \sqrt{2}$ .

**Solution approach B:** Starting from the end, finding  $f_{\min}$  first, the minimal takes place when

$$\begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

is co-linear with  $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ , and in a direction that subtracts. The difference of the two lengths is the solution of the minimization problem, and this is

$$f_{\min} = \sqrt{5} - \sqrt{2}.$$

The angle between the two vectors is

$$\arccos \frac{\langle v_1, v_2 \rangle}{\|v_1\| \cdot \|v_2\|} = \arccos \left( \frac{3}{\sqrt{10}} \right) > 0.$$

This is obtuse, and hence the minimizing value for  $\theta$ , that rotates (clock-wise) and aligns  $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$  in a direction opposite to  $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$  is

$$\theta_{\min} = \pi - \arccos \left( \frac{3}{\sqrt{10}} \right).$$

This is about  $2.8[\text{rad}] = 161[\text{degrees}]$ , in agreement with the first approach.

Workspace for Problem 1: Explain your reasoning/work here.

**Solution:**

**Differential Equations: Problem 2 [10 points]:**

Consider the differential equation

$$\ddot{x} - \dot{x}x(x+1) + x(x+1)(x-1) = 0.$$

Draw the phase portrait (on the  $(x, \dot{x})$ -plane). To this end, you need to:

- i) determine all points of equilibrium (2 points)
- ii) linearized models about the points of equilibrium (2 points)
- iii) identify the nature of the points of equilibrium (3 points)
- iv) draw representative (approximate) trajectories. (3 points)

**Attention:** For one of the points of equilibrium, the linearized model has roots on  $j\mathbb{R}$ . You are not required to carry out higher-order analysis, and you can draw the phase portrait assuming that the trajectories are dictated by the linearized dynamics.

**Solution:**

We first determine a state representation by setting as entries of the state vector  $x_1 = x$  and  $x_2 = \dot{x}$ . The state equations are

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= x_1x_2 + x_1 - x_1^3 + x_1^2x_2\end{aligned}$$

We linearize about points of equilibrium by considering perturbation

$$x_1 = x_{1,\text{eq}} + \xi_1 \text{ and } x_2 = x_{2,\text{eq}} + \xi_2,$$

about  $(x_{1,\text{eq}}, x_{2,\text{eq}})$ , where  $x_{2,\text{eq}} = 0$  always. We thus obtain:

$$\begin{aligned}\dot{\xi}_1 &= \xi_2 \\ \dot{\xi}_2 &= (1 - 3x_{1,\text{eq}}^2)\xi_1 + (x_{1,\text{eq}} + x_{1,\text{eq}}^2)\xi_2.\end{aligned}$$

The points of equilibrium are three:  $x_2 = 0$  while  $x_1 \in \{0, \pm 1\}$ .

Linearization about  $(0, 0)$  gives  $\dot{x}_1 = x_2$  and  $\dot{x}_2 = x_1$ . This is a saddle point, since the linearized system

$$\frac{d}{dt} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix},$$

with eigenvalues  $\pm 1$  for the state matrix and eigenvectors  $(1, 1)^T$  and  $(1, -1)^T$ .

The point of equilibrium at  $(1, 0)$  is an unstable focus, where trajectories rotate clock-wise. The linearization gives

$$\frac{d}{dt} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -2 & 2 \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix},$$

with eigenvalues the roots of  $s^2 - 2s + 2$ , i.e.,  $1 \pm i$ . The clockwise rotation is dictated by the fact that the vector field flows to the right about the  $x_2 = 0$  axis. Finally, the equilibrium at  $(-1, 0)$  has a linearization

$$\frac{d}{dt} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -2 & 0 \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix},$$

with eigenvalues the roots of  $s^2 + 2$ , i.e.,  $\pm i\sqrt{2}$ , which indicates that it is likely a center.

However, higher order analysis is needed to ascertain or not. In this case it is actually a stable focus. You are not required to carry out this.

The phase portrait is shown in the figure.

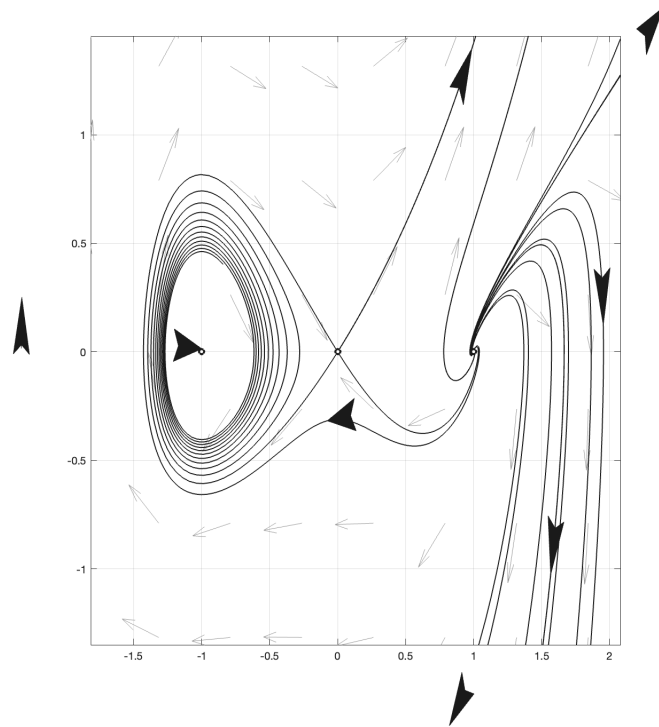


Figure 1: Phase portrait

Workspace for Problem 2: Explain your reasoning/work here.

**Solution:**

**Partial Differential Equations: Problem 3 [10 points]:**

Consider the wave equation

$$u_{tt} - a^2 u_{xx} = 0$$

for the infinite domain  $-\infty < x < +\infty$  and  $t > 0$  with  $u(0, x) = \delta(x)$ , the Dirac delta function, and  $u_t(0, x) = 0$ . Find  $u(t, x)$  by using (1) Fourier transform and (2) D'Alembert traveling wave formula. No need to use tables. Use original definitions to find transforms and inverse transforms if you do not remember a property, a transform, or inverse transform.

Workspace for Problem 3: Explain your reasoning/work here.

**Solution:**

(1) Using Fourier Transform Method

Let us consider the more general initial conditions

$$u(0, x) = f(x)$$

$$u_t(0, x) = 0$$

Define Fourier Transform in  $x$  of  $u(t, x)$  as

$$U(t, \omega) = \mathcal{F}\{u(t, x)\} \equiv \int_{-\infty}^{+\infty} u(t, x) e^{-i\omega x} dx$$

Its inverse transform is then

$$u(t, x) = \mathcal{F}^{-1}\{u(t, \omega)\} \equiv \frac{1}{2\pi} \int_{-\infty}^{+\infty} U(t, \omega) e^{i\omega x} d\omega$$

We then have

$$\mathcal{F}\left\{\frac{d^n u(t, x)}{dx^n}\right\} = (i\omega)^n U(t, \omega)$$

Taking the Fourier transform of the original PDE and initial conditions, we get

$$\frac{d^2 U(t, \omega)}{dt^2} + a^2 \omega^2 U = 0$$

$$U(t=0, \omega) = \mathcal{F}\{f(x)\} = F(\omega); \quad \frac{dU}{dt}(t=0, \omega) = 0$$

The solution is then

$$U(t, \omega) = \mathcal{F}\{f(x)\} \cos(a\omega t) = F(\omega) \frac{e^{-ia\omega t} + e^{ia\omega t}}{2} = \frac{1}{2} F(\omega) e^{-ia\omega t} + \frac{1}{2} F(\omega) e^{ia\omega t}$$

Notice the translation principle for Fourier transform (if you do not remember it, you can easily derive it by the definition of Fourier transform.)

$$\mathcal{F}\{f(x \pm \Delta x)\} = \mathcal{F}\{f\} e^{\pm i\Delta x \omega} = F(\omega) e^{\pm i\Delta x \omega}$$

Thus, we get

$$u(t, x) = \frac{1}{2} f(x - at) + \frac{1}{2} f(x + at)$$

For the present case  $f(x) = \delta(x)$ . Therefore, we have our final solution

$$u(t, x) = \frac{1}{2} \delta(x - at) + \frac{1}{2} \delta(x + at)$$

If you work directly with the original delta function you will easily get from the definitions of Fourier transform and the delta function the following transform pairs.

$$\mathcal{F}\{\delta(x)\} = \int_{-\infty}^{+\infty} \delta(x) e^{-i\omega x} dx = 1$$

and

$$\mathcal{F}\{\delta(x-at)\} = \int_{-\infty}^{+\infty} \delta(x-at)e^{-i\omega x}dx = e^{-i\omega at}$$

The same final answer as above boxed equation will arise.

(2) Using d'Alembert formula

Consider the given wave equations with the general initial conditions:

$$u(0, x) = f(x)$$

$$u_t(0, x) = g(x)$$

D'Alembert obtained the following general solution to the above problem.

$$u(x, t) = \frac{1}{2}[f(x-at) + f(x+at)] + \frac{1}{2a} \int_{x-at}^{x+at} g(\xi)d\xi$$

which immediately yield the above boxed answer. If you do not remember the above d'Alembert formula, you should be able to easily derive it in the exam (at least for the case when  $g(x) = 0$ ) by knowing the general solution to the wave equation  $u_{tt} = a^2 u_{xx}$

$$u(x, t) = \phi(x-at) + \psi(x+at)$$

(You may find the derivation of the above in almost all textbooks on PDEs.)

Workspace for Problem 3: Explain your reasoning/work here.

**Partial Differential Equations: Problem 4 [10 points]:**

Solve the Laplace equation

$$\nabla^2 u(r, \theta) = 0$$

in the circle  $r < R, 0 < \theta < 2\pi$  with the boundary condition  $u(R, \theta) = f(\theta)$ . Express your solution in terms of integral(s) of  $f(\theta)$ .

Workspace for Problem 4: Explain your reasoning/work here.

**Solution:**

In polar coordinate system, the PDE becomes

$$\nabla^2 u = \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial u}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0$$

Let  $u(r, \theta) = g(r)h(\theta)$  and substitute it into the above PDE, we obtain

$$\frac{r \frac{d}{dr} \left( r \frac{d}{dr} \right) g(r)}{g(r)} = -\frac{h''(\theta)}{h(\theta)} = \lambda$$

$\lambda$  above must be a constant since the first term in the above equation depends only on  $r$  while the second term depends only on  $\theta$ . The only possibility for them to be equal is for them to be a constant. Therefore, we obtain two separate equations

$$\begin{aligned} h''(\theta) + \lambda h(\theta) &= 0 \\ r \frac{d}{dr} \left( r \frac{d}{dr} \right) g(r) - \lambda g(r) &= 0 \end{aligned}$$

The first equation poses an eigenvalue problem with periodic boundary conditions in  $\theta$ . For  $\lambda < 0$ , the solution to the ODE is  $h(\theta) = a \cosh(\sqrt{-\lambda}\theta) + b \sinh(\sqrt{-\lambda}\theta)$ , which cannot satisfy the required period boundary conditions. Therefore,  $\lambda < 0$  cannot be an eigenvalue. For  $\lambda \geq 0$ , the solution to the ODE is  $h(\theta) = a \cos(\sqrt{\lambda}\theta) + b \sin(\sqrt{\lambda}\theta)$ , which will satisfy the period boundary condition with period  $2\pi$  if  $\lambda = m^2, m = 0, 1, \dots$ . Therefore, we find the following set of eigenvalues and eigenfunctions.

| m               | eigenvalue $\lambda$ | eigenfunction(s)               |
|-----------------|----------------------|--------------------------------|
| 0               | $\lambda = m^2 = 0$  | $\cos(m\theta) = 1$            |
| $m=1, 2, \dots$ | $\lambda_m = m^2$    | $\cos(m\theta), \sin(m\theta)$ |

Once  $\lambda$  is determined as above as  $\lambda = m^2, m = 0, 2, \dots$ , the ODE for  $g(r)$  becomes

$$r \frac{d}{dr} \left( r \frac{d}{dr} \right) g(r) - m^2 g(r) = 0$$

which is the so-called equi-dimensional equation or Euler's equation, which has the solution form of  $g(r) = r^n$  for  $m > 0$ . Substituting it into the equation, we see  $n$  must be  $n = \pm m$  and

$$g(r) = ar^m + br^{-m}, \quad m = 1, 2, \dots$$

For  $m = 0$ , it is easy to integrate directly and obtain the general solution

$$g(r) = a + b \ln r$$

Since  $g(r)$  must be bounded for any  $r < R$ , we must have  $b = 0$  for all of the above solutions. Therefore, the solution can be expressed as

$$g(r) = r^m$$

for all  $m = 0, 1, 2, \dots$ . We then obtain the following series solution to the PDE problem

$$u(r, \theta) = \sum_{m=0}^{+\infty} [a_m \cos(m\theta) + b_m \sin(m\theta)] r^m$$

Application of the boundary conditions at  $r = R$  gives

$$\begin{aligned} a_0 &= \frac{1}{2\pi} \int_0^{2\pi} f(\theta) d\theta, & b_0 &= 0 \\ a_m &= \frac{1}{\pi R^m} \int_0^{2\pi} f(\theta) \cos(m\theta) d\theta, & b_m &= \frac{1}{\pi R^m} \int_0^{2\pi} f(\theta) \sin(m\theta) d\theta, \quad m = 1, 2, \dots \end{aligned}$$

Workspace for Problem 4: Explain your reasoning/work here.